

IoT PROJECT – REVIEW-I REPORT

1. General Information

Project Title: IoT-Based Automatic Rain Sensing System Using Arduino Nano

Programme: Master of Computer Applications (MCA)

Year: 2

Semester: III

Review: Review-I

Review Dates: 24-08-2026 to 28-08-2026

Team Members:

1. K. Mahitha.
2. J. Ajay
3. Y. Prakash
4. G. Manibabu

Project Guide: Uppalapati Nagaratna Kumari

Project Guide: Ketha Siva Saikishore

Department: Master of Computer Applications

2. Project Abstract

The **IoT-Based Automatic Rain Sensing System Using Arduino Nano** is designed to automatically detect rainfall and operate a protective shade. The system uses a rain sensor to detect water droplets and an Arduino Nano to process the sensor signal. When rain is detected, the Arduino Nano sends a control signal to a servo motor. The servo motor automatically moves the protective shade into the required position to protect clothes or outdoor objects from rain. When the rain stops, the shade can return to its original position. The proposed system reduces manual effort and provides a simple, low-cost automation solution. The project demonstrates the integration of a rain sensor, Arduino Nano, and servo motor for automatic rain-based protection.

3. Introduction

Rain is a natural environmental condition that can cause problems when clothes and other objects are kept outdoors. Normally, people need to manually observe the weather and move the clothes when rain starts. This can be difficult when the user is away or when rain starts suddenly.

The proposed **Automatic Rain Sensing System Using Arduino Nano** system provides an automatic solution for this problem. A rain sensor is placed in an appropriate outdoor position to detect the presence of rain. The sensor sends its reading to the Arduino Nano, which acts as the main controller.

When rain is detected, the Arduino Nano controls a servo motor to automatically move the protective shade. The shade covers the required area and protects the clothes or objects from getting wet. When

the rain condition stops, the shade can be returned to its original position. Thus, the system reduces manual effort and provides automatic rain-based protection.

4. Problem Statement

Outdoor clothes and objects can become wet when rain starts unexpectedly. Manual monitoring and operation of a protective shade require continuous attention from the user.

Major problems identified:

- Difficulty in detecting sudden rainfall.
- Need for continuous manual observation.
- Clothes kept outdoors may get wet.
- Manual operation of the protective shade is inconvenient.
- No automatic mechanism to respond to rainfall.
- The user may not be present when rain starts.

Therefore, an automatic rain sensing and shade-control system is proposed to detect rain and automatically operate the protective shade.

5. Project Objectives

The main objectives of the project are:

1. To develop an automatic rain sensing system using Arduino Nano.
2. To detect rainfall using a rain sensor.
3. To process the rain sensor readings using Arduino Nano.
4. To automatically control a servo motor when rain is detected.
5. To move the protective shade into position during rainfall.
6. To protect clothes and outdoor objects from rain.
7. To reduce the need for manual operation.

To develop a simple and low-cost automatic protection system.

6. Proposed IoT Modules

Module 1 – Rain Detection

A rain sensor is used to detect water droplets and identify the presence of rainfall.

Module 2 – Arduino Nano Controller

The Arduino Nano receives the sensor reading and processes it to determine the rain condition.

Module 3 – Rain Condition Decision

The Arduino compares the sensor value with a predefined threshold to determine whether rain is detected.

Module 4 – Servo Motor Control

When rain is detected, Arduino Nano sends a control signal to the servo motor.

Module 5 – Automatic Shade Mechanism

The servo motor moves the shade automatically to cover the required area and protect the clothes.

Module 6 – Shade Return Mechanism

When the rain condition stops, the servo motor can move the shade back to its original position.

Module 7 – Future IoT Extension

The current system mainly performs local automatic control. In the future, an IoT/Wi-Fi module can be added to monitor the rain condition remotely and provide notifications if required.

7. Hardware Requirements

S.No.	Component	Purpose
1	Arduino Nano	Main controller
2	Rain Sensor	Detects rainfall
3	Servo Motor	Controls shade movement
4	Protective Shade	Protects clothes/objects from rain
5	Breadboard	Circuit development
6	Jumper Wires	Electrical connections
7	USB Cable	Programming and power
8	Power Supply	Provides power to the system

8. Software Requirements

- Arduino IDE
- Embedded C/C++ programming
- Arduino Nano board support
- Servo motor library

- Serial Monitor
- USB drivers, if required

9. Input and Expected Output

Inputs

The system receives:

- Rain sensor readings
- Water droplets/rainfall
- Sensor analog or digital signal
- Rain threshold value

Expected Outputs

The system produces:

- Detection of rain
- Automatic servo motor movement
- Automatic movement/folding of the protective shade
- Protection of clothes or outdoor objects

Return of the shade to its original position when appropriate

10. Algorithm

1. Start the system.
2. Initialize Arduino Nano.
3. Initialize the rain sensor.
4. Initialize the servo motor.
5. Read the rain sensor value.
6. Compare the sensor value with the predefined threshold.
7. If rain is detected, send a control signal to the servo motor.
8. Move the servo motor to operate the protective shade.
9. Keep the shade in the protective position while rain is detected.
10. When rain stops, move the servo motor back to the required original position.
11. Continue monitoring the rain sensor.
12. Stop the system when required.

11. System Flow

Rain → Rain Sensor → Arduino Nano → Decision → Servo Motor → Automatic Shade

The system continuously monitors the surrounding environment using a rain sensor. When rain droplets are detected, the sensor sends the signal to the Arduino Nano. The Arduino processes the sensor reading and determines the rain condition.

If rain is detected, the Arduino Nano sends a control signal to the servo motor. The servo motor automatically moves the protective shade to cover the clothes or required area. When there is no rain, the shade remains in its normal position.

12. Flowchart

START



Initialize Arduino



Initialize Rain Sensor



Initialize Servo Motor



Read Rain Sensor



Is Rain Detected?

YES → Activate Servo → Move Protective Shade → Protect Clothes



Continue Monitoring

NO → Keep Shade Normal/Open



Continue Monitoring



END

13. Implementation Progress

Activity	Status
Project topic selection	Completed
Problem identification	Completed
Background study	Completed
Project objectives	Completed
Hardware identification	Completed
Software identification	Completed
System architecture	Completed
Algorithm	Completed
Flowchart	Completed
Rain sensor interfacing	In Progress / Completed
Arduino Nano programming	In Progress
Servo motor interfacing	In Progress / Completed
Shade mechanism	In Progress
Rain detection testing	In Progress
Automatic shade operation	In Progress
IoT connectivity	Planned
Complete prototype	Planned

14. Completed Module Demonstration

Demonstration 1 – Rain Sensor

The rain sensor is connected to the Arduino Nano. When water droplets fall on the sensor, the sensor value changes.

Demonstration 2 – Sensor Reading

The Arduino Nano reads and processes the rain sensor value through the program.

Demonstration 3 – Servo Motor

When rain is detected, Arduino Nano sends a signal to the servo motor.

Demonstration 4 – Automatic Shade

The servo motor moves the protective shade automatically to cover the clothes or required area.

15. Sample Output

No Rain

Automatic Rain Sensing System

Rain Sensor : Normal

Rain Status : NO RAIN

Shade : OPEN

Servo : 0°

Rain Detected

Automatic Rain Sensing System

Rain Sensor : Wet

Rain Status : RAIN DETECTED

Shade : CLOSED

Servo : 90°

16. Individual Team Member Contribution

Team Member	Contribution
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Mahitha.Kari	Problem identification, background study and documentation
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Ajay.Jonnada	Hardware selection and rain sensor interfacing
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Prakash.Yaramalla	Arduino Nano programming, algorithm and servo control
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Manibabu.Gopiseti	Shade mechanism, testing, integration and presentation
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Each team member should explain their individual contribution during the Review-I presentation.

17. Challenges Identified

The team has identified the following technical challenges:

1. Accurate detection of rainfall.

2. Proper rain sensor calibration.
3. Selecting a suitable rain detection threshold.
4. Correct servo motor movement.
5. Designing a suitable protective shade mechanism.
6. Integrating the rain sensor, Arduino Nano and servo motor.
7. Ensuring reliable automatic operation.
8. Establishing IoT/Wi-Fi connectivity for future implementation.

18. Proposed Solutions

The team proposes to address these challenges by:

- Calibrating the rain sensor before final implementation.
- Testing the sensor under different rainfall conditions.
- Using suitable threshold values for rain detection.
- Testing different servo motor angles.
- Designing and testing the protective shade mechanism.
- Implementing proper control logic in Arduino Nano.
- Performing repeated hardware and software integration tests.
- Testing Wi-Fi connectivity when the IoT module is added.

19. Work Planned After Review-I

1. Complete rain sensor calibration.
2. Complete Arduino Nano programming.
3. Complete servo motor interfacing.
4. Improve the automatic shade mechanism.
5. Implement accurate rain detection logic.
6. Test the system under different rain conditions.
7. Add IoT/Wi-Fi connectivity.
8. Implement remote monitoring and notification.
9. Integrate all modules.
10. Conduct complete system testing.

11. Prepare the final working prototype.
12. Complete final project documentation.

20. Conclusion

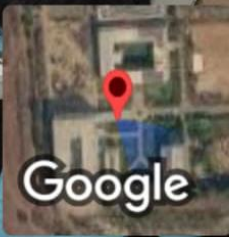
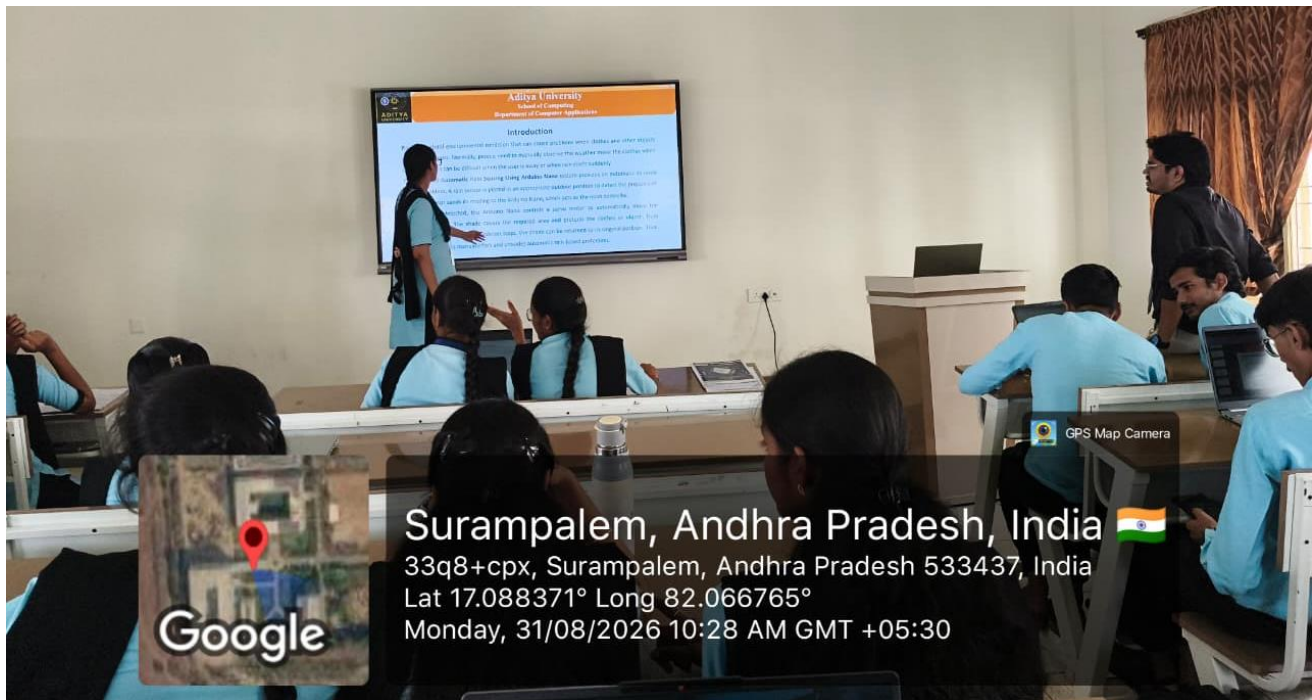
The **IoT-Based Automatic Rain Sensing Using Arduino Nano** project provides an automatic solution for protecting clothes and outdoor objects from unexpected rainfall. The rain sensor detects the presence of water, and the Arduino Nano processes the sensor information. When rain is detected, the Arduino controls a servo motor that automatically operates the protective shade.

The Review-I phase has established the project problem, objectives, hardware and software requirements, system flow, algorithm, and initial implementation. The next stage will focus on completing the servo-controlled shade mechanism, calibrating the rain sensor, testing the complete system, and improving the reliability of automatic operation.

The project demonstrates how a sensor, microcontroller, and actuator can work together to create a simple and useful automation system.

21. Review-I Presentation Checklist

- Individual contributions explained
- Project abstract submitted
- Problem statement explained
- Objectives explained
- IoT modules/functions explained
- Hardware and software requirements explained
- Inputs and outputs explained
- Algorithm explained
- Flowchart presented
- System architecture presented
- Individual responsibilities explained
- Completed modules demonstrated
- Code/functions explained
- Sample output demonstrated
- IoT concepts explained
- Progress status presented
- Future work explained



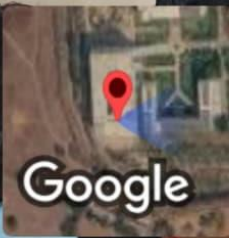
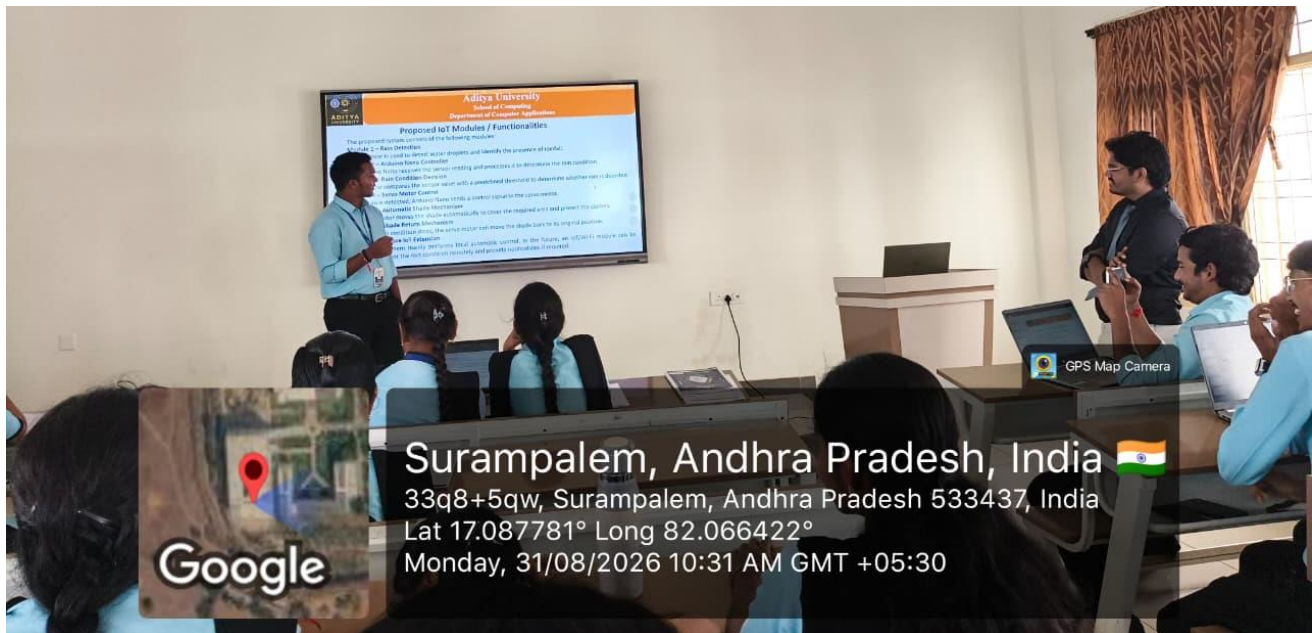
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